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Operator Projection for Depth-Robust Optimization in Factored Networks



(/pdf?)

id=VWKMApNcbq)

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04 May 2026 (modified: 27 May 2026) NeurIPS 2026 Conference Submission

Conference, Senior Area Chairs, Area Chairs, Reviewers, Authors

Revisions (/revisions?id=VWKMApNcbq)

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Abstract:

Parameter-space optimization can be a poor proxy for function-space optimization in deep factored networks. We study this gap through the composed operator $P = W_L \cdots W_1$ and show that standard gradient updates of the factors do not realize the operator gradient step: each layer contributes a spectral distortion, producing a depth-dependent mismatch between the intended and actual motion of P . We propose to correct this mismatch directly via an operator-projection objective, which searches for factor updates whose product best matches a target operator-space update. In deep linear networks, this objective admits an Alternating Least Squares (ALS) solver with an explicit modewise form; this view also reveals K-FAC as a separable approximation to the operator-projection update. Since random initialization can independently cause spectral collapse of the context operators, we introduce a warm-start that preserves their singular values and makes the ALS sub-problems well conditioned. Across controlled linear, fixed-gate, and non-linear experiments, ALS with warm-start achieves accurate operator alignment and remains stable at depths where standard parameter-space optimizers collapse. These results support operator projection as a principled framework for diagnosing and correcting depth-induced optimization mismatch in factored networks.

Checklist Confirmation: I confirm that I have included a paper checklist in the paper PDF.

Responsible Reviewing: We acknowledge the responsible reviewing obligations as authors.

Primary Area: Optimization (e.g., convex and non-convex, stochastic, robust)

Secondary Area: Theory (e.g., learning theory, theory of deep learning, algorithmic game theory)

Contribution Type: General: Most submissions will fall into this type.

Academic Integrity: I acknowledge that I have read the NeurIPS Handbook and commit to adhering to all policies in the Handbook (<https://neurips.cc/Conferences/2026/MainTrackHandbook> (<https://neurips.cc/Conferences/2026/MainTrackHandbook>)), the NeurIPS Code of Conduct and the NeurIPS Academic Integrity Policy.

LLM Usage: Editing (e.g., grammar, spelling, word choice)

Declaration: I confirm that the above information is accurate.

Reviewer Nomination: Augustin M Cosse (/profile?id=~Augustin_M_Cosse1)

Submission Number: 27284

[Discussion \(?id=VWKMApNcbq#discussion\)](/forum?id=VWKMApNcbq#discussion)

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Citation Checker by Program Chairs

Citation Checker by Program Chairs  04 Aug 2026, 14:26 (modified: 05 Sept 2026, 01:18) Program Chairs, Senior Area Chairs, Area Chairs, Authors  Revisions (/revisions?id=81XXLvRFY)

Comment:

Automatically flagged references

Here we list the references that were automatically flagged by our checker. Note that our checker may miss many hallucinated references, and some of the listed references may have been flagged erroneously. **It is therefore crucial that you check them manually, following the guidance above. We rely on the Area Chairs to decide whether these constitute violations of responsible authorship.**

For papers falling into either of the following categories, we expect your default recommendation to be a rejection on the grounds of irresponsible authorship. This is strong presumption rather than a strict rule; as the Area Chair, you retain discretion, and may recommend otherwise where the specific circumstances of a paper warrant it:

1. The paper contains at least two hallucinated references according to the criteria described below the report.
2. The paper contains one hallucinated reference and there is some additional evidence of AI slop.

If a paper has only one problematic reference, but no additional concerns about the quality of the paper or its presentation, please simply alert the authors in your meta-review.

Please complete the form "Hallucinated References Investigation" to communicate the results of your investigation prior to completing your meta reviews.

Citation Checker Report

Reference Check Summary

5/33 problematic references.

List of Problematic References with Reasoning

```
@article{agarwal2026newton,  
  author = {Agarwal, Naman and Cesista, Franz Louis and others},  
  title = {The newton-muon optimizer},  
  journal = {arXiv preprint arXiv:2604.01472},  
  year = {2026}  
}
```

The paper titled 'The newton-muon optimizer' with arXiv ID 2604.01472 exists, but its actual authors are Zhehang Du and Weijie Su. The citation incorrectly attributes the paper to Naman Agarwal and Franz Louis Cesista.

```
@inproceedings{arora2018optimization,
  author = {Sanjeev Arora and Nadav Cohen and Noah Golowich and Alexander Rakhlin},
  title = {On the optimization of deep networks: Implicit acceleration by overparameterization},
  booktitle = {International Conference on Machine Learning},
  pages = {244--253},
  year = {2018}
}
```

The paper 'On the optimization of deep networks: Implicit acceleration by overparameterization' (ICML 2018) was authored by Sanjeev Arora, Nadav Cohen, and Elad Hazan. The provided BibTeX entry incorrectly replaces Elad Hazan with Noah Golowich and Alexander Rakhlin.

```
@inproceedings{gunasekar2017implicit,
  author = {Suriya Gunasekar and Jason Lee and Daniel Soudry and Nathan Srebro},
  title = {Implicit regularization in matrix factorization},
  booktitle = {Advances in Neural Information Processing Systems},
  volume = {30},
  year = {2017}
}
```

The paper 'Implicit regularization in matrix factorization' exists and was published at NeurIPS 2017 (Advances in Neural Information Processing Systems 30). However, the author list in the citation is incorrect. The actual authors are Suriya Gunasekar, Blake Woodworth, Srinadh Bhojanapalli, Behnam Neyshabur, and Nathan Srebro. The provided citation incorrectly replaces Blake Woodworth, Srinadh Bhojanapalli, and Behnam Neyshabur with Jason Lee and Daniel Soudry (who were co-authors on a different paper by Gunasekar and Srebro).

```
@inproceedings{kim2026convergence,
  author = {Joonwon Kim and Sangmin Oh},
  title = {Convergence of muon with newton-schulz},
  booktitle = {International Conference on Learning Representations},
  year = {2026}
}
```

Although the paper 'Convergence of Muon with Newton-Schulz' exists and was published at ICLR 2026, the author names in the citation ('Joonwon Kim and Sangmin Oh') are incorrect. The actual authors of the paper are Gyu Yeol Kim and Min-hwan Oh.

```
@article{kosson2024spectral,
  author = {Atli Kosson and Thijs Vogels and Martin Jaggi},
  title = {Spectral preconditioning for gradient methods on graded non-convex functions},
  journal = {arXiv preprint arXiv:2408.05044},
  year = {2024}
}
```

Although a paper titled 'Spectral preconditioning for gradient methods on graded non-convex functions' exists, the provided citation contains critical errors in the metadata. The actual authors are Nikita Doikov, Sebastian U. Stich, and Martin Jaggi (not Atli Kosson and Thijs Vogels), and the correct arXiv identifier is 2402.04843 (arXiv:2408.05044 refers to an unrelated optics paper).

Hallucinated/problematic reference criteria

These are adopted from ICLR and ICML 2026 hallucinated reference guidance:

- If you are unable to find any document or publication with a title anywhere near the one provided in the

reference, it should be considered hallucinated. Minor differences in title should not be considered evidence of a hallucinated reference.

- If the reference's authors are only slightly incorrect, e.g., a few subtle misspellings, missing a few authors, or wrong authorship order, we do not consider this to imply a hallucinated reference, but please let the submission's authors know in your meta-review that they should fix this issue.
- If the author list has additional issues, e.g., added authors that are not actually authors of that publication or missing a large fraction of authors, then the reference should be considered hallucinated.
- If the reference's venue is a real venue but is incorrect for that reference, we do not consider this to imply a hallucinated reference, but please tell the submission's authors in your meta-review, these include wrong journal name, arXiv ID, etc.
- If there is no evidence that the reference's venue exists, the reference should be considered hallucinated.
- Finally, we note that there are many generic references that refer to existing artifacts (e.g. "GPT-4o, OpenAI et al. 2024") but do not refer to any real document. These are fine.
- Even with the above guidelines in mind, there will likely be some borderline cases. Please bring those up with your SAC if you need additional guidance.

Official Review of Submission27284 by Reviewer NGj5

Official Review by Reviewer NGj5 📅 29 Jun 2026, 04:00 (modified: 04 Aug 2026, 09:51)

👁 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer NGj5

📄 Revisions (/revisions?id=5PpjgHJW9y)

Summary:

The paper studies why parameter-space gradient descent is a poor proxy for function-space (operator) optimization in deep factored networks. For a composed operator $P = W_L \cdots W_1$, the authors show that simultaneous first-order GD on the factors does **not** realize the clean operator step $- \eta G$. Instead, defining context matrices $A_k = W_L \cdots W_{k+1}$ and $B_k = W_{k-1} \cdots W_1$, the induced increment is $\Delta P_{GD} = - \eta \sum_k A_k A_k^T G B_k^T B_k$ (Prop. 3.1) — a left/right preconditioned, depth-accumulating distortion of the intended motion.

To fix this they propose an **operator-projection** objective: find factor increments $\{\Delta W_\ell\}$ minimizing $\|\Pi(W_\ell + \Delta W_\ell) - P_{tgt}\|^2 + \lambda \sum \|\Delta W_\ell\|^2$ with $P_{tgt} = P - \eta G$. For deep linear nets this is solved by Alternating Least Squares (ALS): each layer's sub-problem is a Tikhonov-regularized LS that, via compact SVDs of A_k, B_k , decouples into a closed-form **modewise** filter $\Delta \tilde{W}_{\{k, ij\}} = \tilde{G}_{\{k, ij\}} / (\sigma_{\{A, i\}}^2 \sigma_{\{B, j\}}^2 + \lambda)$. They show K-FAC is the **separable** denominator $(\sigma_{A^2} + \lambda)(\sigma_{B^2} + \lambda)$ under isotropic data — a clean, falsifiable connection. Because random init causes spectral collapse, they add an identity/orthogonal **warm-start** that restores unit context singular values. The formulation extends to fixed-gate linear nets (FGLN, Eq. 9) and frozen-gate ReLU MLPs (Eqs. 10–13), the latter giving a sum-of-Kroneckers system solved exactly (ALS-Exact, $O(L \cdot d^6)$) or approximately (ALS-MN, $O(L \cdot d^3)$). Experiments: a 2D toy, deep linear/FGLN targets up to $L=128$, and MNIST MLPs up to $L=32$, where ALS+warm-start stays trainable while baselines collapse.

Contribution Type: Negative Results: The main contribution is in understanding a negative result. (The significance and originality bar for these contributions is high.)

Strengths And Weaknesses:

Strengths

1. **Correct, well-posed diagnostic.** Prop. 3.1 cleanly exposes the depth-dependent operator distortion as a sum of context-Gram preconditioners. The exact-agreement condition $\sum_k A_k A_k^T \otimes B_k^T B_k = I$ is a crisp way to state why GD cannot track $- \eta G$.
2. **The K-FAC link (Sec. 3.3) is the real contribution.** Deriving K-FAC's update as the separable denominator $(\sigma_{A^2} + \lambda)(\sigma_{B^2} + \lambda)$ vs the exact $\sigma_{A^2} \sigma_{B^2} + \lambda$, both in basis (V_A, U_B) , is informative and testable. The non-zero gap "grows with λ " is a concrete, checkable claim.
3. **Commendable candor.** They state ALS is not a practical optimizer, MNIST is not a benchmark, the MLP warm-start has no guarantee, and the deep-linear identity warm-start is "essentially an artefact." This honesty raises trust.
4. **Thoughtful ablation.** Separating Xavier (ill-conditioned) from identity (well-conditioned) init to disentangle *spectral collapse* from *direction mismatch* (Fig. 5, "identity-init recovers flat spectrum") is the

right experiment.

Weaknesses

1. **Novelty is thin / mostly reframing.** Operator vs parameter-space dynamics in deep linear nets is well-trodden (Saxe 2014; Arora 2018/2019; Gunasekar 2017; Du 2018). Casting the GD step as LS projection onto P_{tgt} and solving by ALS repackages known matrix-factorization geometry. The K-FAC observation is the only new edge, and only under isotropy.
2. **Solver acknowledged impractical.** ALS-Exact MLP is $O(L \cdot d^6)$; ALS-MN is $O(L \cdot d^3)$; baselines are $O(L \cdot d^2)$. Width is capped at 32 to stay tractable. "Depth-robust *optimization*" overstates it — this is a diagnostic, not an optimizer. No wall-clock numbers anywhere despite the orders-of-magnitude cost gap.
3. **Unfair comparison.** ALS uses **lr=1.0** while baselines use **lr=1e-3** (toy) or a coarse $\{1e-2 \dots 1e-5\}$ sweep (MNIST). Such a learning-rate disparity alone can produce the cosine table (Table 1: ALS 1.00 vs HeavyBall 0.90, K-FAC 0.97). Without matched per-step budgets and wall-clock, "baselines collapse" is not isolated to direction mismatch.
4. **Headline result is confounded.** The $L=128$, $d=16$ Xavier win (ALS hits 2×10^{-10} at step 677; baselines stuck at ≈ 0.97) blends direction mismatch with vanishing-gradient collapse. The cleaner identity-init regime removes collapse but, per Remark 4.1, with $\lambda=0$ the first sweep absorbs the whole update in one layer — "trivial." The truly non-trivial $\lambda>0$ case is never isolated as the headline.
5. **Toy scale, no real baselines-as-intended.** $d=16$ linear, bias-free width-32 MNIST, single GPU. No CIFAR, no conv/transformer, no batchnorm/bias, no comparison vs depth-aware LR/warmup or μP . The frozen-gate surrogate is untested beyond MNIST and gates are *frozen*, sidestepping the actual nonlinear difficulty.

References

- A. M. Saxe, J. L. McClelland, S. Ganguli. *Exact solutions to the nonlinear dynamics of learning in deep linear neural networks*. ICLR 2014.
- S. Arora, N. Cohen, E. Hazan. *On the Optimization of Deep Networks: Implicit Acceleration by Overparameterization*. ICML 2018.
- S. S. Du, W. Hu, J. D. Lee. *Algorithmic Regularization in Learning Deep Homogeneous Models: Layers are Automatically Balanced*. NeurIPS 2018.
- S. Gunasekar, B. Woodworth, S. Bhojanapalli, B. Neyshabur, N. Srebro. *Implicit Regularization in Matrix Factorization*. NeurIPS 2017.
- S. Arora, N. Cohen, W. Hu, Y. Luo. *Implicit Regularization in Deep Matrix Factorization*. NeurIPS 2019.

Quality: 3: good

Clarity: 2: not good

Significance: 2: not good

Originality: 2: not good

Questions:

- **Matched budgets:** Per unit wall-clock and with tuned learning rates, how does ALS-MN compare to Muon/Shampoo/SOAP? Table 1 reflects an lr disparity (1.0 vs 1e-3).
- **Non-trivial regime:** In the identity-init experiment, is the gap robust to $\lambda>0$, or does it shrink toward the Remark 4.1 trivial case?
- **K-FAC claim:** Can you empirically show the $(\sigma^2 + \lambda)(\sigma^2 + \lambda)$ vs $\sigma^2 \sigma^2 + \lambda$ gap predicts K-FAC error as data departs isotropy?
- **Beyond toy:** Does the advantage survive bias, BatchNorm, width ≥ 256 , and learned (not frozen) gates?

Limitations:

yes

Confidence: 3: You are fairly confident in your assessment. It is possible that you did not understand some parts of the submission or that you are unfamiliar with some pieces of related work. Math/other details were not carefully checked.

Ethical Concerns: NO or VERY MINOR ethics concerns only

Paper Formatting Concerns:

No noticeable formatting issues.

Code Of Conduct Acknowledgement: Yes

Responsible Reviewing Acknowledgement: Yes



Rebuttal by Authors

Rebuttal

by Authors (👤 Jed A. Duersch (/profile?id=~Jed_A._Duersch1), Zied Bouraoui (/profile?id=~Zied_Bouraoui1), Augustin Cosse (/profile?id=~Augustin_Cosse1), Nathanaël Haas (/profile?id=~Nathanaël_Haas1))

📅 28 Jul 2026, 02:33 (modified: 28 Jul 2026, 10:16)

👁 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors

📄 Revisions (/revisions?id=WhEfQovFaA)

Rebuttal:

We thank the reviewer for the detailed reading.

On novelty, we agree that the descriptive geometry of factorized gradient descent is well established, and we will present Proposition 3.1 as setup. Our contribution is the inverse construction: specify a desired operator displacement and solve for factor increments that realize it. To our knowledge, objective (3), its modewise block solution (6), and the resulting comparison with K-FAC have not previously been formulated. We will avoid broader claims about discovering the underlying spectral bias.

On cost and intended use, we agree that the present solver is impractical and will reframe the paper as a diagnostic contribution. Its purpose is to compare parameter-space optimization with an operator-aligned control, not to claim deployment-level superiority. We add wall-clock and peak-memory results. At $d = 16$, $L = 128$ in float64, ALS-exact costs 104 ms/step versus 2.9 ms for Adam on CPU and 683 ms versus 4.0 ms on GPU. Peak GPU memory at this size is 0.9MB for ALS-exact versus 1.5MB for Adam and up to 3.6MB for K-FAC; at $d = 256$, $L = 32$, ALS-exact is the lowest of all seven methods at 26.8MB against 64–89MB for the adaptive baselines, since it does not retain per-layer state across the full depth (full table under Reviewer 4jZv, point on cost).

On the learning-rate comparison, η defines the operator target $P_{\text{tgt}} = P - \eta G$ for ALS rather than directly scaling a parameter update. The realized operator-step norms in Table 1 are also of the same order for ALS and most baselines. More directly, the cosine in Table 1 is learning-rate invariant to first order because the positive scalar η cancels after normalization. Since the exact product update contains higher-order terms, we also re-measured the cosine over learning rates from 10^{-6} to 10^{-1} . It varies by at most 10^{-3} for Heavy Ball, Adam, Muon, Shampoo, and SOAP. K-FAC is equally stable for $\eta \leq 10^{-3}$ and changes only at larger steps, outside the local regime; Table 1 uses 5×10^{-5} . We additionally use 11 log-spaced learning rates over this range and a $4 \times$ longer training horizon; every baseline optimum lies inside the grid.

On the warm-start confound and the non-trivial $\lambda > 0$ regime, we agree that the original presentation conflated two effects. The Xavier experiment shows recovery from an initially collapsed context and therefore evaluates the combined ALS-plus-warm-start method. The identity experiment uses no warm-start and isolates the update mismatch. We repeated the latter at $d = 16$, $L = 128$ for $\lambda \in 0, 10^{-6}, 10^{-4}, 10^{-2}$. The final relative operator error is 7.7×10^{-8} across this range. For $\lambda > 0$, the increment is distributed across all 128 layers (participation ratio 128), so the result does not rely on the $\lambda = 0$ case in Remark 4.1, where one layer can absorb the entire update.

On K-FAC away from isotropy, we ran the requested test with $d = 16$, $L \in 8, 32$, population covariances, $\Sigma_g = I$, and Σ_x varying from isotropy to condition number 10^4 (log-spaced, trace normalized), for $\lambda \in 10^{-4}, 10^{-2}, 10^{-1}$. At isotropy, the measured denominator difference agrees with the expression in Eq. (8) to relative error below 10^{-7} . On Haar-orthogonal weight states, the per-layer relative deviation of the K-FAC update from the exact modewise solve increases to 0.83–0.90 at condition number 10^4 . In the ALS basis, the denominator-only discrepancy is 0.51–0.65, while the eigenbasis misalignment of the K-FAC input factor with U_B reaches 0.71. These measurements quantify both parts of the stated limitation: away from isotropy, K-FAC differs in both denominator and basis.

On scale, architectures, and frozen gates, we agree that the current experiments do not establish a large-scale practical benefit: the $O(Ld^6)$ exact MLP solve precludes such a study. The frozen-gate

construction is a local linearization, not a claim to solve the full nonlinear problem exactly. Gates are frozen within each outer step and recomputed between steps; Section 5.3 therefore trains actual ReLU MLPs, albeit only on MNIST at width 32. Convolutional networks, transformers, biases, and BatchNorm remain outside the present empirical scope, and we will state this explicitly. We also add $\eta/\sqrt{L}$, η/L , and linear-warmup baselines for Adam and Heavy Ball. None escapes the Xavier collapse at $L = 128$ (final errors 0.93--0.97), and none removes the identity-initialized collapse-and-recovery transient.



Reply to the authors

Official Comment by Reviewer NGj5 📅 02 Aug 2026, 04:40

👁 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors

Comment:

Thank you for the thorough and candid rebuttal. I appreciate the new wall-clock/peak-memory numbers, the learning-rate-invariance analysis of the cosine metric together with the 11-point LR sweep and longer horizon, the identity-init $\lambda > 0$ experiment isolating the update mismatch, and the K-FAC anisotropy study at condition number $1e4$. These directly address my concerns on fairness (W3), the confounded $L=128$ headline (W4), missing cost numbers (W2), and my K-FAC question (Q3), and I have updated my review to reflect that.

To be transparent about how I weigh the response as a whole: the clarifications resolve the specific issues I raised, but they do not address the two factors that most drive my rating: the incremental novelty with respect to the operator-vs-parameter-space literature, and the limited, toy-scale evaluation (frozen gates, width-32 MNIST; Q4 left out of scope). As these remain open, I am maintaining my score while raising my confidence.



➡ Replying to Reply to the authors

Official Comment by Authors

Official Comment

by Authors (👁 Jed A. Duersch (/profile?id=~Jed_A._Duersch1), Zied Bouraoui (/profile?id=~Zied_Bouraoui1), Augustin Cosse (/profile?id=~Augustin_Cosse1), Nathanaël Haas (/profile?id=~Nathanaël_Haas1))

📅 02 Aug 2026, 15:34

👁 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors

Comment:

We appreciate your confirmation that the **rebuttal resolves the concerns** about **computational cost, tuning fairness, the confounded headline experiment, the non-trivial $\lambda > 0$ regime**, and the **K-FAC prediction**.

We would like to clarify why these two remaining concerns do not diminish the contribution, since the novelty lies in a new operator-space intervention and the experimental scale is consistent with the paper's diagnostic objective.

On novelty, it is clear the forward problem is known: prior work characterizes the operator dynamics, balancedness, and implicit regularization induced by gradient descent on factors. Our contribution is the inverse problem: prescribe an operator displacement and solve for factor updates that realize it. This is an intervention rather than another description of the same dynamics. It produces objective (3), the modewise block solution (6), the precise K-FAC comparison, and, most importantly, a counterfactual control that can suppress the additional distortion induced by factorization while retaining the same factorized representation. To our knowledge, the cited operator-dynamics and implicit-bias literature does not provide such a control.

Regarding experimental scale, the paper does not claim that the current ALS solver is a scalable replacement for existing optimizers. Its purpose is to provide an exact diagnostic with which the operator trajectory, context spectra, and competing optimization effects

can be measured separately. The controlled linear and frozen-gate settings are therefore aligned with the paper's central claim. Experiments on substantially larger architectures would test the scalability of the current solver, which we do not claim, rather than the validity or usefulness of the diagnostic itself.

As detailed in our response to Reviewer U6A5, we have also run a concrete implicit-bias experiment that demonstrates this use of the diagnostic. In an underdetermined rank-2 matrix-sensing problem, at matched training MSE, factorized Adam obtains held-out MSE 7.7×10^{-3} and effective rank 5.7, whereas direct- P Adam obtains 4.9×10^{-2} and effective rank 14.9. Operator projection uses the same three-factor parameterization and initialization as factorized Adam, but tracks direct- P Adam to relative error 4.0×10^{-6} and follows its higher-rank, worse-generalizing trajectory. This isolates a regime in which the additional bias induced by factorized optimization is beneficial. It also demonstrates the concrete role of the diagnostic: the contribution of the factorization bias becomes measurable through a matched counterfactual rather than being inferred only from the solution reached by factorized training.

We therefore believe that the inverse optimization primitive, its falsifiable connection to K-FAC, and this controlled implicit-bias intervention establish a substantive theoretical and diagnostic contribution within the paper's stated scope. We would be grateful if you would reconsider whether the revised evidence and clarified contribution affect your overall assessment.

Official Review of Submission27284 by Reviewer U6A5

Official Review by Reviewer U6A5 📅 26 Jun 2026, 08:09 (modified: 07 Aug 2026, 12:16)

👁 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer U6A5

📄 Revisions (/revisions?id=THayzogP1C)

Summary:

The paper points out that updating weight parameters of deep networks using gradient based methods is not the same as doing the gradient step in the operator-space. They propose a method to fix this.

Contribution Type: General: Most submissions will fall into this type.

Strengths And Weaknesses:

The paper is well-written and highlights an important property of current training methods of deep neural networks. However, it is already known that weight-space updates are not equivalent to operator space updates. It is known that this leads to a spectral bias. The authors acknowledge that the "fix" they propose may not be practical. That alone does not mean it is not interesting from a conceptual point of view. But what I feel is missing from the paper is a justification that such a fix is in fact desirable. Maybe the gradient descent updates in the weight space result in a favorable implicit bias? Or maybe not. That would possibly be the more interesting question to address before considering alternatives to the standard gradient based weight updates.

Quality: 3: good

Clarity: 4: excellent

Significance: 2: not good

Originality: 2: not good

Questions:

Can you provide comprehensive experiments that confirm that the networks obtained when training with your method are performing better in some practical sense than networks trained in a traditional way (e.g. better generalization)?

Limitations:

Yes

Confidence: 4: You are confident in your assessment, but not absolutely certain. It is unlikely, but not impossible, that you did not understand some parts of the submission or that you are unfamiliar with some pieces of related work.

Ethical Concerns: NO or VERY MINOR ethics concerns only

Paper Formatting Concerns:

None

Code Of Conduct Acknowledgement: Yes

Responsible Reviewing Acknowledgement: Yes



Rebuttal by Authors

Rebuttal

by Authors ( Jed A. Duersch (/profile?id=~Jed_A._Duersch1), Zied Bouraoui (/profile?id=~Zied_Bouraoui1), Augustin Cosse (/profile?id=~Augustin_Cosse1), Nathanaël Haas (/profile?id=~Nathanaël_Haas1))

 28 Jul 2026, 02:34 (modified: 28 Jul 2026, 10:16)

 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors

 Revisions (/revisions?id=LscvNOmRUv)

Rebuttal:

We thank the reviewer for identifying the central interpretive question.

We agree that the non-equivalence of weight-space and operator-space updates is known, especially for deep linear networks. We will present Proposition 3.1 as setup and narrow the novelty claim to the operator-projection objective, its modewise ALS block solve, and the K-FAC comparison.

We also agree that operator alignment is not automatically desirable. The implicit bias of parameter-space training may improve generalization in some regimes. Our claim is narrower: the projection supplies a control condition that explicitly corrects the factorization-induced distortion, allowing its optimization effect to be measured rather than assumed.

The identity-initialized experiment gives one such measurement. Although every context singular value initially equals 1, all tested parameter-space methods transiently collapse the tail of the operator spectrum, whereas the projected update follows the target trajectory without this detour. The recovery time grows with dimension and depth and is learning-rate sensitive. At $d = 64$, Adam at the learning rate selected on a 2,000-step horizon reduces the operator flatness to 2×10^{-6} and remains above 10^{-1} relative error after 20,000 steps; K-FAC and Shampoo, whose updates have two-sided structure, recover almost immediately. Thus, in this controlled regime, factorization bias imposes a measurable optimization cost even from a perfectly conditioned start. We do not infer from this that the bias is harmful in every regime.

Regarding practical performance, Figure 6 reports validation loss over a learning-rate sweep and three seeds. The operator-projected method remains trainable at $L = 32$, where the tested gradient-based baselines reach chance-level loss. We will add test accuracy for transparency. However, these results demonstrate optimization and trainability at depth, not improved generalization when both approaches train successfully. A comprehensive generalization comparison at realistic scale would require a substantially more efficient projection solver. We therefore retitle and reframe the paper as a theoretical diagnostic and explicitly leave that question open.



Official Comment by Reviewer U6A5

Official Comment by Reviewer U6A5  31 Jul 2026, 01:12

 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors

Comment:

Thank you for your response. I think I now understand your intended contribution much better! And I very much appreciate your point that you may be able to improve trainability. I find the paper interesting overall, but, at least at this stage of the work, remain skeptical that its significance is high enough. However, I remain open-minded. It would be nice to have some clear, concrete, ideally easy to understand, use-case. What can we now understand that we did not understand before? How would I make use of the diagnostic? How can it improve anything I am doing? Can we investigate the implicit bias

implications?



➔ *Replying to Official Comment by Reviewer U6A5*

Official Comment by Authors

Official Comment

by Authors (👤 Jed A. Duersch (/profile?id=~Jed_A._Duersch1), Zied Bouraoui (/profile?id=~Zied_Bouraoui1), Augustin Cosse (/profile?id=~Augustin_Cosse1), Nathanaël Haas (/profile?id=~Nathanaël_Haas1))

📅 02 Aug 2026, 04:17

👤 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors

Comment:

We are grateful for this, below we give what the diagnostic makes visible that was not previously measurable, three specific ways it can be used, and a direct measurement on the implicit-bias question.

We can now understand two things better with this framework.

The first is that the spectral bias at depth is not only an asymptotic preference among minimizers. It is an active transient. Starting from identity initialization, where every context singular value equals 1 and the operator spectrum is exactly flat, every tested parameter-space optimizer destroys that conditioning before recovering it. The tail of the operator spectrum collapses and then has to be rebuilt. Existing descriptive analyses characterize where factorized gradient descent ends up but does not show this transient and sensitive collapse. This can be attributed to the implicit bias of gradient descent in factored models and is not a necessity of training itself.

The second is that the framework predicts which optimizer families avoid the detour, and the prediction holds. K-FAC and Shampoo, whose updates carry two-sided Kronecker structure, recover almost immediately, whereas Adam, Muon and SOAP do not. Section 3.3 gives the mechanism: in the isotropic deep linear case, the K-FAC denominator $(\sigma_A^2 + \lambda)(\sigma_B^2 + \lambda)$ approximates the exact $\sigma_A^2 \sigma_B^2 + \lambda$ in the same basis (V_A, U_B) , so K-FAC is partially operator aligned by construction, while elementwise or one-sided methods are not. The common observation that two-sided preconditioners hold up better at depth acquires an explanation and a quantitative form.

We can do diagnostics in 3 ways.

First, as a measurement, without running ALS at all. For any factored model and any optimizer, the contexts A_k and B_k are computable and Proposition 3.1 gives the realized operator increment. The cosine between that increment and $-\eta G$ is a cheap per-step scalar, one $O(Ld^3)$ pass, and allows to compute the operator misalignment. This in turn could help diagnose training characteristics in several settings, allowing to attribute the role of implicit bias to different effects, as well as to understand better training dynamics and the effects of hyperparameters on operator alignment, stalling loss etc. ALS is needed only to produce the reference trajectory (compute the weights updates realizing the reference operator step), not to take the measurement.

Second, as a design criterion for preconditioners and their damping. Equation (8) states exactly what an operator-aligned denominator would be and how far a separable one departs from it, as a function of λ and the context spectra. The gap $\lambda(\sigma_{A,i}^2 + \sigma_{B,j}^2 + \lambda - 1)$ grows with λ , so the departure of K-FAC from operator alignment is worst precisely in the high-damping regime that is often adopted for stability at depth. This is directly actionable, and Appendix B.3 describes the corresponding variant that keeps the K-FAC cost structure while replacing the separable denominator with the exact one.

Third, as a control condition for experiments about the bias itself, which is your last question.



Official Comment by Authors

Official Comment

by Authors (👤 Jed A. Duersch (/profile?id=~Jed_A._Duersch1), Zied Bouraoui (/profile?id=~Zied_Bouraoui1), Augustin Cosse (/profile?id=~Augustin_Cosse1), Nathanaël Haas (/profile?id=~Nathanaël_Haas1))

📅 02 Aug 2026, 04:17

👤 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors

Comment:

On investigating the implicit bias.

We agree that this is one of the most interesting questions, and we ran the following controlled experiment. In noiseless dense teacher-student regression with $n \geq d$ and a full-rank design matrix, the observations uniquely determine the operator, leaving no ambiguity among interpolating solutions. We therefore use an underdetermined matrix-sensing problem following Gunasekar et al. (2017). The target is a rank-2 operator $P^* \in \mathbb{R}^{20 \times 20}$, and the training set contains $m = 160$ random Gaussian measurements $y_s = \langle M_s, P^* \rangle$. Since $m < d^2 = 400$, these measurements do not uniquely determine an unrestricted operator, and many operators interpolate them exactly.

All methods start from the same small composed operator $P_0 = 0.01I$. For the factorized methods, we use the balanced initialization $W_1 = W_2 = W_3 = 0.01^{1/3}I$. We compare: (a) Adam applied to the factors of $P = W_3 W_2 W_1$; (b) Adam applied directly to P , which removes the additional bias introduced by factorization while retaining Adam's own implicit bias; and (c) operator projection using the same factorization as (a). At each step of (c), the target operator update is proposed by a shadow direct- P Adam optimizer evaluated at the current composed operator, with identical hyperparameters and persistent first- and second-moment states. Operator projection then solves for factor updates that realize this unfactorized target.

At matched training MSE (8.8×10^{-9} for all three methods), factorized Adam obtains a held-out measurement MSE of 7.7×10^{-3} and relative recovery error $|P - P^*|_F / |P^*|_F = 0.34$. Its spectrum is substantially more concentrated: 12 of 20 singular values exceed 1 of the largest, and its entropy effective rank is 5.7, compared with the target's effective rank of 2. Direct Adam obtains a held-out MSE of 4.9×10^{-2} , $6.4 \times$ higher, and relative recovery error 0.90. It retains 19 populated singular values and has effective rank 14.9.

Operator projection closely tracks direct- P Adam, reaching the matched stopping point with $|P_{\text{proj}} - P_{\text{direct}}|_F / |P_{\text{direct}}|_F = 4.0 \times 10^{-6}$. Correspondingly, it follows the unfactorized method's generalization behavior and full-rank spectral trajectory rather than the more concentrated spectrum selected by factorized Adam. Because operator projection and factorized Adam use the same parameterization and initialization, this comparison isolates the difference in their update dynamics rather than their representational capacity. In this experiment, projection suppresses the factorization-induced trajectory distortion to numerical precision.

These results are consistent with the classical prediction that factorized optimization can favor lower-rank solutions and that this preference can improve generalization in an underdetermined problem. We do not conclude that the factorization bias is universally beneficial. Rather, the experiment demonstrates how operator projection can serve as a control that switches off the additional factorization-induced bias while leaving the factorized representation unchanged. Here, the controlled comparison shows that the bias is beneficial; in another regime, the conclusion could differ. We will add the singular-value trajectories and complete experimental protocol to the appendix.

Official Review of Submission27284 by Reviewer 4jZv

Official Review by Reviewer 4jZv 📅 25 Jun 2026, 23:24 (modified: 13 Aug 2026, 07:29)

👁 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer 4jZv

📄 Revisions (/revisions?id=5UyayVcBBJ)

Summary:

The paper under review identifies a deviation between the parameter-wise gradient descent and the operator gradient descent on factored networks $P = W_L \cdot W_1$. It proves that there is a spectrally filtered operator distortion that accumulates as depth grows. As a cure, the authors propose an algorithmic improvement by applying a operator-projection solved by Alternating Least Squares (ALS). This approach has to be implemented with a warm-start initialization to avoid spectral collapse of context operators. The new algorithm was tested with 2D operator trajectories, and on deep linear and fixed-gate linear networks in teacher-student settings, and on a frozen-gate MNIST MLP.

Contribution Type: General: Most submissions will fall into this type.

Strengths And Weaknesses:

Strength:

For depth-induced optimization and factorized training, the paper reveals a critical caveat. Showing modern preconditioned optimizers fail on a simple teacher-student problem when depth is large is a noteworthy phenomenon. This observation is validated in experiments, where even with perfect conditioning, classical gradient descent methods still stall over an error level, which was then broken by the proposed ALS descent.

Weaknesses

- In generic form, the ALS descent cost is $O(Ld^6)$ and completely impractical. This is intrinsic to modewise solves with full context SVDs per layer per sweep. Even in good special cases, the cost is still $O(Ld^3)$ and much higher than classical descent.
- The ALS descent is not robust as a method. There is no theoretical guarantee for convergence of the ALS descent and warm start is necessary. It would be helpful to have a condition/monitoring mechanism against collapsing instead of relying on specialized initialization (identity / isometric FGLN / batch-orthogonal MLP heuristic without formal guarantee).
- I believe the claim that the minimizer of Eq. (3) is unique for all L is only layer wise true but doesn't hold globally. The method is solving a intrinsically highly non-convex optimization problem, which explains the lack of robustness.

Quality: 3: good

Clarity: 3: good

Significance: 2: not good

Originality: 3: good

Questions:

- Could you quantify the attribution to the success of experiments between warm start and operator-projection?
- You claimed that "the minimizer of (3) exists and is unique for all L ". Could you provide a proof?

Limitations:

Mostly yes. One major limitation of the ALS descent is the lack of robustness in generic settings, which is partially acknowledged in the paper.

Confidence: 4: You are confident in your assessment, but not absolutely certain. It is unlikely, but not impossible, that you did not understand some parts of the submission or that you are unfamiliar with some pieces of related work.

Ethical Concerns: NO or VERY MINOR ethics concerns only

Paper Formatting Concerns:

none

Code Of Conduct Acknowledgement: Yes

Responsible Reviewing Acknowledgement: Yes



Rebuttal by Authors

Rebuttal

by Authors (👤 Jed A. Duersch (/profile?id=~Jed_A._Duersch1), Zied Bouraoui (/profile?id=~Zied_Bouraoui1), Augustin Cosse (/profile?id=~Augustin_Cosse1), Nathanaël Haas (/profile?id=~Nathanaël_Haas1))

📅 28 Jul 2026, 02:34 (modified: 28 Jul 2026, 10:16)

👁 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors

📄 Revisions (/revisions?id=6HZOk2MKmX)

Rebuttal:

We thank the reviewer for pinpointing the global-uniqueness error and the distinction between the projection objective and its solver.

On cost, we agree that the $O(Ld^6)$ exact MLP solve and the $O(Ld^3)$ special cases are far more expensive than standard $O(Ld^2)$ optimizers. We will present the method as a diagnostic rather than a practical optimizer and add wall-clock and peak-memory measurements. At $d = 16$, $L = 128$ in float64, ALS-exact costs 104 ms/step versus 2.9 ms for Adam on CPU and 683 ms versus 4.0 ms on GPU; at $d = 256$, $L = 32$, an ALS step reaches 1.0 s.

Full peak GPU memory table (CUDA, float64, MB):

d	L	Heavy Ball	Adam	Muon	K-FAC	Shampoo	SOAP	ALS-exact
16	32	8.5	0.4	0.3	1.2	0.9	0.9	0.7
64	32	2.1	4.0	2.3	7.9	5.2	6.3	2.4
256	32	32.1	64.0	36.0	79.8	71.8	88.8	26.8
16	128	1.3	1.5	1.3	3.6	1.8	2.0	0.9

ALS-exact's peak memory grows far more slowly with d and L , becoming the lowest of all seven methods at the two larger configurations tested ($d = 256$ and $L = 128$), while at the smallest scale ($d = 16$, $L = 32$) its transient solve cost exceeds the state cost of the lightest baselines (Adam, Muon).

On existence, uniqueness, and convergence, the reviewer is correct: the global uniqueness statement in Appendix A is false. For $\lambda > 0$, objective (3) is continuous and coercive, so a global minimizer exists, but it need not be unique. For example, with scalar factors, $L = 2$, $W_1 = W_2 = 0$, and $t > \lambda > 0$, the objective $(\Delta w_1 \Delta w_2 - t)^2 + \lambda(\Delta w_1^2 + \Delta w_2^2)$ has the two global minimizers $\Delta w_1 = \Delta w_2 = \pm\sqrt{t - \lambda}$. Only the block minimizer is unique: for $\lambda > 0$, subproblem (4) is 2λ -strongly convex and has solution (6).

For cyclic exact block minimization with a fixed regularization anchor, objective (3) is non-increasing, its values converge, and coercivity keeps the iterates bounded. Since the objective is continuously differentiable and each block minimizer is unique, every accumulation point is stationary by the standard block-coordinate-descent result (Bertsekas, Proposition 2.7.1). We claim neither convergence to a global minimizer nor numerical robustness in collapsed contexts. In float64 tests over 72 identity, Xavier, and Ginibre instances, each well-conditioned run was monotone to roundoff; the failures occurred when context products underflowed or the solve became numerically ill-conditioned.

On attribution, the warm-start initializes the increments used by the ALS solve. It is not an initialization supplied to the baseline networks, but it materially affects the first ALS step and should be treated as part of the method. Under Xavier initialization, the contexts are collapsed before optimization begins. The current ALS solver, like the baselines, fails without the warm-start. Under exact identity initialization, no warm-start is used. With sufficiently conservative per-method tuning, the baselines can eventually converge, but they take a collapse-and-recovery detour whose duration grows with d and L ; ALS avoids this detour.

Starting instead from $W_\ell = 0.99I$ tests sensitivity to a small per-layer perturbation. Under the tested protocol, relative errors are 3.6×10^{-3} and 2.1×10^{-3} at $L = 128, 256$, but 35 and 79 at $L = 512, 1024$. The initial product has already contracted to $0.99^{512} \approx 6 \times 10^{-3}$ and $0.99^{1024} \approx 3 \times 10^{-5}$. ALS-exact remains stable at $L = 512, 1024$ without a warm-start. We have

not exhaustively swept learning rates for this ablation, so we present it as evidence of sensitivity rather than an impossibility result.

These experiments support a three-regime interpretation: exact identity is an unusually favorable special case; a small per-layer deviation can compound into poor end-to-end conditioning at sufficient depth; and Xavier initialization is already severely collapsed, which is where the warm-start is needed. The projection benefit persists in the second regime with the warm-start disabled, while the third regime exposes a limitation of the current solver.

On a collapse monitor, Equation (6) gives a direct diagnostic. For each layer, we track $\rho_k = \left(\min_{i \in \mathcal{I}_k} \sigma_{A,i}^2 \right) \left(\min_{j \in \mathcal{J}_k} \sigma_{B,j}^2 \right) / \lambda$, where $\mathcal{I}_k$ and $\mathcal{J}_k$ exclude only structural zero modes imposed by the architecture or gates. Values $\rho_k \ll 1$ indicate that the denominator is dominated by λ and the update has lost its operator-correcting structure. At $d = 16$, $L = 128$, and $\lambda = 10^{-4}$, Xavier initialization without warm-start flags all 128 layer solves, with context-Gram products underflowing to zero; identity initialization and the warm-start flag none, with products near 1. We will add this monitor to the released deep-linear and MLP implementations and describe it in Section 4. We also explain why reducing λ alone does not restore missing context directions: it rescales the update but does not improve the underlying context geometry.

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